

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

1. A and B are two events such that $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cup B) = 0.65$. Which of the following statements is true:
 - a. A and B are mutually exclusive events.
 - b. A and B are dependent events.
 - c. A and B are impossible events.
 - d. A and B are independent events.

2. A set of data has no outliers. The sample mean is 154, while the sample median is 120. The distribution of the sample data is:
 - a. fairly symmetric
 - b. right skewed
 - c. left skewed

3. People who are easily recruited are used in the sampling method called:
 - a. Simple random sample.
 - b. Stratified random sampling.
 - c. Convenience sampling.
 - d. Cluster sampling.

4. If you toss a fair coin and roll a fair die.
 - a. Define the sample space?
 - b. Define event A as getting tails. List the outcomes in A and find $P(A)$.
 - c. Define event B as getting two heads. List the outcomes in B and find $P(B)$.
 - d. Define event C as rolling at least a 4. List the outcomes in C and find $P(C)$.
 - e. Are A and C mutually exclusive events? Are they independent events?

5. A manufacturer of calculators buys integrated circuits from supplies A, B, and C. Fifty percent of the circuits come from A, 30% from B, and 20% from C. One percent of the circuits supplied by A have been defective in the past, 3% of B's have been defective, and 4% of C's have been defective. A circuit is selected at random and found to be defective. What is the probability it was manufactured by B?

6. The table below gives information on the type of coffee selected by someone purchasing a single cup at a particular airport kiosk.

	Small	Medium	Large
Regular	14%	20%	26%
Decaf	20%	10%	10%

Consider randomly selecting a coffee purchaser.

- What is the probability that the individual purchased a cup of decaf?
- If we learn that the selected individual purchased a small cup, what is the probability that he chose decaf coffee?
- If we learn that the selected individual purchased decaf coffee, what is the probability that he chose a small size?

7. Suppose 2% of cotton fabric rolls contain flaws and 3% of nylon fabric rolls contain flaws. Of the rolls used by a manufacturer, 70% are cotton and 30% are nylon.

- What is probability that a randomly selected roll used by the manufacturer contains flaws?
- If a randomly selected roll contains flaws, what is probability that it is cotton?

8. An educator wants to compare the effectiveness of computer software that teaches reading with that of a standard reading curriculum. He tests the reading ability of each student in a class of fourth graders, then divides them into two groups. One group uses the computer regularly, while the other studies a standard curriculum. At the end of the year, he retests all the students and compares the increase in reading ability in the two groups. Is this an experiment or an observational study? Explain.

9. What term would best describe the shape of the given boxplot:

- Symmetric
- Right (positive) skewed.
- Left (negative) skewed.



10. At a certain restaurant, they are trying two new items. The first is a main entrée (*Dish 1*) and the second is a side dish (*Dish 2*). Tracking orders has shown for each customer, $P(\text{Dish 1}) = 0.15$, $P(\text{Dish 2}) = .30$, and $P(\text{Dish 1} \cap \text{Dish 2}) = 0.045$. Calculate the following:

a) $P(\text{Dish 1} \cup \text{Dish 2})$

b) $P(\text{Dish 1}')$ (this is the complement of *Dish 1*)

11. Consider randomly selecting a student at a certain university, and let A denote the event that the selected individual has a Visa credit card and B be the analogous event for a MasterCard. Suppose that $P(A)=0.5$, $P(B)=0.4$, and $P(A \cap B)=0.06$.

a) Compute the probability that a selected individual has both a Visa credit card or a MasterCard.

b) Compute the probability that a selected individual does not have a Visa card.

12. An arena in a large city is used for several different types of events. One event can be held at a time. The table shows the events at the arena the next several months. A randomly chosen day with an event occurring is selected and the event is happening is recorded.

Event	Percent
Basketball	24%
Ice hockey	23%
Concert	16%
Ice show	11%
Trade show	8%
Other	18%

a) Will this result in mutually exclusive events? Explain.

b) What is the probability that a randomly chosen event is an ice hockey game or an ice show?

13. Tell whether the events are mutually exclusive.

a) **Event A:** A student knows how to play a musical instrument.

Event B: A student doesn't know how to play a musical instrument.

b) **Event A:** A student knows how to play the clarinet.

Event B: A student knows how to play the guitar.

14. Using #13 part b above:

a) What does $A \cup B$ mean in words? Use what the events mean in words.

b) What does B' mean in words? Use what the events mean in words.

c) What does $A' \cap B$ mean in words? Use what the events mean in words.

15. A box contains 4 bolts. Two of them, labeled #1 and #2, are 5 mm in diameter, and two of them, labeled #3 and #4, are 7 mm in diameter. Bolts are randomly selected until a 5 mm bolt is obtained.

a) Give the sample space.

b) Let A be the event that only one bolt is selected. Find A.

c) Let B be the event that three bolts are selected. Find B.